Persephone:
The Mathematics of Persistent Homology for Images

A rigorous companion to the Persephone visualization studio.
Everything on screen, derived from first principles, with full proofs.

Scope. Persephone treats a grayscale image as a discrete Morse function and animates the persistent homology of its sublevel-set filtration. This note develops every concept that appears on the screen (cubical complexes, simplicial homology, persistence modules, the structure theorem, persistence diagrams, barcodes, the Bubenik landscape, Alexander duality, and the union–find algorithm that drives it all) with proofs given in full whenever the technique is elementary and as detailed sketches when the proof is long but standard, with precise pointers to the canonical sources [EH10, Car09, Bub15, CSEH07, Hat02, ZC05, CB15].

Contents

1 Preliminaries and notation **1**

2 Cubical complexes on the pixel grid **2**

3 Discrete Morse functions and the filtration **3**

4 Cellular homology over $\mathbb{F}_2$ **3**

5 Persistent homology **5**
5.1 The structure theorem in finite type 5
5.2 Persistence diagram and barcode 6

6 Computation: union–find with the elder rule **7**

7 Alexander duality and H_1 for images **8**

8 Persistence landscapes **10**

9 Stability **11**

10 What every panel shows **13**

References **14**

1 Preliminaries and notation

We work over the finite field $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ throughout. Homology with $\mathbb{F}_2$ coefficients suffices for everything in the visualization and lets us treat the persistence module as a $\mathbb{F}_2[t]$-module with no signs or orientation issues. Throughout, *vector space* means $\mathbb{F}_2$-*vector space* and *linear* means $\mathbb{F}_2$-*linear*, unless we explicitly say otherwise.

Definition 1.1 (Image as a scalar field)**.** A grayscale image of width w and height h is a function

$$f : \Omega \to \mathbb{R}, \qquad \Omega := \{0, 1, \ldots, w-1\} \times \{0, 1, \ldots, h-1\} \subset \mathbb{Z}^2.$$

We call Ω the *pixel grid* and f a *discrete scalar field*.

Definition 1.2 (Sublevel and superlevel sets)**.** For $t \in \mathbb{R}$, the *sublevel set* at parameter t is

$$\Omega_{\leq t} \;:=\; \{p \in \Omega \mid f(p) \leq t\},$$

and the *superlevel set* is $\Omega_{>t} := \{p \in \Omega \mid f(p) > t\}$.

The family $\{\Omega_{\leq t}\}_{t\in\mathbb{R}}$ is monotone: $s \leq t$ implies $\Omega_{\leq s} \subseteq \Omega_{\leq t}$. This monotone family is the central object of study, the *sublevel-set filtration*.

2 Cubical complexes on the pixel grid

To do topology on a pixel grid we need a combinatorial model of its geometric realization. The cleanest is the *cubical complex*.

Definition 2.1 (Elementary cube)**.** An *elementary interval* is a subset of $\mathbb{R}$ of the form $[k, k]$ (a point, length 0) or $[k, k+1]$ (a unit segment, length 1) for some $k \in \mathbb{Z}$. An *elementary cube* of dimension d is a Cartesian product $Q = I_1 \times \cdots \times I_n \subset \mathbb{R}^n$ of elementary intervals where exactly d of the I_j are unit segments and the rest are degenerate.

Definition 2.2 (Cubical complex of the pixel grid)**.** Let $\Omega = \{0, \ldots, w-1\} \times \{0, \ldots, h-1\}$. The cubical complex X associated to Ω places one vertex at every pixel centre:

- 0*-cells (vertices):* the points $p = (i, j) \in \Omega$, one per pixel;
- 1*-cells (edges):* the unit segments $[i, i+1] \times \{j\}$ and $\{i\} \times [j, j+1]$ joining two 4-adjacent pixels;
- 2*-cells (squares):* the unit squares $[i, i+1] \times [j, j+1]$ whose four corners are pixels.

The geometric realization $|X|$ is the rectangle $R := [0, w-1] \times [0, h-1] \subset \mathbb{R}^2$.

The V-construction. The filtration value of a cell must be compatible with the function f on pixels. We use the *V-construction* (cf. [WCV12]), in which pixels are vertices: every cell σ is assigned

$$F(\sigma) \;:=\; \max\{f(p) : p \in \Omega \text{ a vertex of } \sigma\}.$$

Concretely: a vertex p has $F(p) = f(p)$; an edge between 4-adjacent pixels p and q has $F = \max(f(p), f(q))$; a square takes the maximum over its four corner pixels.

Lemma 2.3 (Cubical sublevel filtration)**.** Define $X_t := \{\sigma \in X : F(\sigma) \leq t\}$. Then $\{X_t\}_{t\in\mathbb{R}}$ is a monotone family of cubical subcomplexes (each X_t closed under faces, and $s \leq t \Rightarrow X_s \subseteq X_t$). Concretely, X_t is the full subcomplex spanned by $\Omega_{\leq t}$: its vertices are the pixels with $f \leq t$, its edges join 4-adjacent such pixels, and its squares are the unit squares whose four corners all have $f \leq t$.

Proof. The vertices of a face of σ are vertices of σ, so $F(\tau) \leq F(\sigma)$ whenever $\tau \subseteq \sigma$; hence X_t is closed under faces. Monotonicity is immediate. Finally $F(\sigma) \leq t$ iff every vertex of σ lies in $\Omega_{\leq t}$, which is the description of the full subcomplex. □

Remark 2.4 (Connectivity convention)**.** In X_t two pixels that touch only diagonally are *not* joined: the diagonal of a unit square is not a cell, and the square itself enters only when all four corners do. Components of X_t are therefore the 4-connected components of $\Omega_{\le t}$, and (Prop. 7.3) the complement $\mathbb{R}^2 \setminus X_t$ is organised by 8-adjacency. This 4/8 pair is the complementary one for which Alexander duality is exact on the pixel grid. The other common model, the T-construction, takes pixels as closed unit squares and the union of the sublevel squares; there two diagonal pixels meet at a corner, which gives the opposite 8/4 convention. Persephone uses the V-construction throughout.

We henceforth use Lemma 2.3 as the definition of the sublevel-set cubical filtration, and conflate "the cubical sublevel set at t" with X_t.

3 Discrete Morse functions and the filtration

Definition 3.1 (Generic function)**.** We say $f : \Omega \to \mathbb{R}$ is *generic* if $f(p) \neq f(q)$ for all $p \neq q$.

Proposition 3.2 (Stepwise growth)**.** If f is generic, the filtration $\{X_t\}_{t\in\mathbb{R}}$ takes finitely many distinct values, and one obtains all of them as

$$\emptyset \;=\; K_0 \;\subsetneq\; K_1 \;\subsetneq\; \dots \;\subsetneq\; K_N \;=\; X, \qquad N = wh,$$

where $K_k = X_{f(p_k)}$ and $p_1, \dots, p_N$ enumerate Ω in increasing order of f. Each step $K_{k-1} \subsetneq K_k$ adds the vertex p_k together with the edges and squares that have p_k as a vertex and all of whose other vertices are already in K_{k-1}; the added cells form the *lower star* of p_k in the V-construction.

Proof. By Lemma 2.3, X_t depends on t only through the set $\Omega_{\le t}$; this set changes exactly at the values $f(p_1) < \dots < f(p_N)$. Between consecutive values $f(p_{k-1})$ and $f(p_k)$ no pixel enters, and the pixel set is fixed at $\{p_1, \dots, p_{k-1}\}$. At $t = f(p_k)$ the pixel p_k is added; the cells σ newly entering X_t are exactly those with p_k as a vertex and all other vertices among $p_1, \dots, p_{k-1}$. This is the lower star of p_k. □

Persephone breaks ties by lexicographic pixel index, so an arbitrary input is reduced to the generic case in a canonical way.

4 Cellular homology over $\mathbb{F}_2$

We recall the algebraic backbone. Let K be a finite CW complex; the cubical complexes above are a special case.

Definition 4.1 (Chain groups)**.** For each integer $k \ge 0$, the k-th *chain group* is the $\mathbb{F}_2$-vector space

$$C_k(K) \;:=\; \mathbb{F}_2\langle \sigma : \sigma \text{ is a } k\text{-cell of } K\rangle,$$

i.e. the free $\mathbb{F}_2$-vector space on the k-cells.

Definition 4.2 (Boundary operator)**.** The *boundary* $\partial_k : C_k \to C_{k-1}$ is the linear map sending a k-cell σ to the formal sum (in $\mathbb{F}_2$) of its codimension-1 faces. For our cubical complex:

$$\begin{aligned}\partial_2\big([i,i{+}1]\times[j,j{+}1]\big) &= [i,i{+}1]\times\{j\} + [i,i{+}1]\times\{j{+}1\}\\ &\quad + \{i\}\times[j,j{+}1] + \{i{+}1\}\times[j,j{+}1],\\ \partial_1\big([i,i{+}1]\times\{j\}\big) &= (i,j)+(i{+}1,j), \qquad \partial_1\big(\{i\}\times[j,j{+}1]\big) = (i,j)+(i,j{+}1).\end{aligned}$$

By convention $\partial_0 = 0$.

Lemma 4.3 ($\partial^2 = 0$)**.** For every $k \geq 1$, $\partial_{k-1} \circ \partial_k = 0$.

Proof. By linearity it suffices to verify the identity on a single k-cell σ. Apply the definition twice:

$$\partial_{k-1}\partial_k(\sigma) \;=\; \sum_{\tau:\, \tau \text{ is a face of } \sigma} \;\sum_{\rho:\, \rho \text{ is a face of } \tau} \rho,$$

where the sums are over codimension-1 faces. Exchange the order of summation: we are counting the multiplicity of each ρ as a codimension-2 face of σ, which is the number of codimension-1 faces τ of σ that simultaneously have ρ as a face.

For any cubical (or simplicial) k-cell σ and any codimension-2 face ρ, there are exactly two codimension-1 faces τ_1, τ_2 of σ with $\rho \subseteq \tau_i$. (This is sometimes called the *diamond property*: each codimension-2 face sits in exactly two codimension-1 faces of the enclosing cell.) For a cube $\sigma = I_1 \times \cdots \times I_d$ this is direct: drop two coordinates to get ρ, then putting either of those two coordinates back as a degenerate point gives the two τ_i.

Hence each ρ appears with coefficient $2 \equiv 0 \pmod 2$, so $\partial_{k-1}\partial_k(\sigma) = 0$. □

Definition 4.4 (Cycles, boundaries, homology)**.**

$$\begin{aligned} Z_k(K) &:= \ker \partial_k && (k\text{-cycles}) \\ B_k(K) &:= \operatorname{im} \partial_{k+1} && (k\text{-boundaries}) \\ H_k(K) &:= Z_k(K)/B_k(K) && (k\text{-th homology}). \end{aligned}$$

By Lemma 4.3, $B_k \subseteq Z_k$, so the quotient makes sense. The k-th *Betti number* is $\beta_k(K) := \dim_{\mathbb{F}_2} H_k(K)$.

Remark 4.5 (Intuitive meaning)**.** For a 2-dimensional cubical complex K:

- $\beta_0(K)$ is the number of path-connected components of $|K|$.
- $\beta_1(K)$ is the number of independent 1-dimensional "holes" (loops that cannot be filled in).
- $\beta_k(K) = 0$ for $k \geq 2$ (the geometric realization sits in $\mathbb{R}^2$).

The first claim is the standard identification of H_0 with the free vector space on components, proved by exhibiting the augmentation $C_0 \to \mathbb{F}_2^{\#\text{components}}$ whose kernel is exactly B_0. The third is dimensional: $C_k = 0$ for $k \geq 3$ in a 2-dimensional cubical complex.

Lemma 4.6 (Rank–nullity for chain groups)**.** For any linear map $\phi : V \to W$ between finite-dimensional $\mathbb{F}_2$-vector spaces, $\dim V = \dim \ker \phi + \dim \operatorname{im} \phi$. In particular, $\dim C_k = \dim Z_k + \dim \operatorname{im} \partial_k$.

Proof. Standard; e.g. choose a basis of $\ker \phi$, extend to a basis of V, and check that ϕ restricted to the complement is an isomorphism onto $\operatorname{im} \phi$. □

Theorem 4.7 (Euler–Poincaré)**.** For a finite CW complex K,

$$\chi(K) \;:=\; \sum_{k \geq 0} (-1)^k \#\{k\text{-cells of } K\} \;=\; \sum_{k \geq 0} (-1)^k \beta_k(K).$$

Proof. Write $c_k := \dim C_k = \#\{k\text{-cells}\}$, $z_k := \dim Z_k$, $b_k := \dim B_k$. By Lemma 4.6 applied to $\partial_k : C_k \to C_{k-1}$,

$$c_k \;=\; z_k + \dim \operatorname{im} \partial_k \;=\; z_k + b_{k-1}.$$

Also $\beta_k = z_k - b_k$ by definition of H_k. Therefore

$$\sum_{k\geq 0}(-1)^k c_k = \sum_{k\geq 0}(-1)^k(z_k + b_{k-1}) = \sum_{k\geq 0}(-1)^k z_k + \sum_{k\geq 0}(-1)^k b_{k-1}.$$

Reindex the second sum by $k \mapsto k+1$: with $b_{-1} := 0$,

$$\sum_{k\geq 0}(-1)^k b_{k-1} \;=\; -\sum_{k\geq 0}(-1)^k b_k.$$

Hence $\sum_k(-1)^k c_k = \sum_k(-1)^k(z_k - b_k) = \sum_k(-1)^k\beta_k$. □

For our cubical sublevel sets this reduces to $\chi(X_t) = \beta_0(X_t) - \beta_1(X_t)$, which is exactly the quantity χ displayed in the invariants panel of the studio.

5 Persistent homology

Homology of a single space discards too much information. The point of persistence is to track *how* cycles appear and disappear along a filtration.

Definition 5.1 (Persistence module)**.** A *persistence module* (over $\mathbb{R}$, indexed pointwise) is a family $\mathbb{V} = \{V_t\}_{t\in\mathbb{R}}$ of $\mathbb{F}_2$-vector spaces together with linear maps $v_{s,t} : V_s \to V_t$ for every $s \leq t$, satisfying

$$v_{t,t} = \mathrm{id}, \qquad v_{s,u} = v_{t,u} \circ v_{s,t} \quad \forall\, s \leq t \leq u.$$

$\mathbb{V}$ is *pointwise finite-dimensional (pfd)* if every V_t has finite dimension. It is of *finite type* if it has only finitely many distinct values: there exist $t_1 < \cdots < t_N$ such that $v_{s,t}$ is an isomorphism whenever s, t lie in the same connected component of $\mathbb{R} \setminus \{t_1, \ldots, t_N\}$.

Proposition 5.2 (Persistent homology is functorial)**.** For a monotone family $\{K_t\}_{t\in\mathbb{R}}$ of finite subcomplexes of a finite CW complex K (i.e. $s \leq t \Rightarrow K_s \subseteq K_t$), the family $H_k(K_\bullet) := \{H_k(K_t)\}_{t\in\mathbb{R}}$ together with the maps $H_k(\iota_{s,t})$ induced by the inclusions $\iota_{s,t} : K_s \hookrightarrow K_t$ is a persistence module of finite type.

Proof. The functoriality of H_k (a cellular inclusion induces a chain map, which induces a map on homology) gives $v_{s,t} := H_k(\iota_{s,t})$. Identity at $s = t$ and composition $v_{s,u} = v_{t,u} \circ v_{s,t}$ follow from the analogous properties of $\iota_{s,t}$. Finite type follows because the filtration changes only at the finitely many values $f(p_1) < \cdots < f(p_N)$ where pixels enter (Prop. 3.2). □

Definition 5.3 (Interval module)**.** For an interval $J = [b, d) \subseteq \mathbb{R}$ (with $-\infty < b < d \leq \infty$), the *interval module* $\mathbb{F}_2[J]$ is the persistence module

$$V_t = \begin{cases} \mathbb{F}_2 & t \in J, \\ 0 & \text{else}, \end{cases} \qquad v_{s,t} = \mathrm{id} \text{ whenever } s, t \in J, \text{ else } 0.$$

5.1 The structure theorem in finite type

Theorem 5.4 (Structure theorem, finite-type version)**.** Let $\mathbb{V}$ be a persistence module of finite type, pfd. Then $\mathbb{V}$ decomposes, uniquely up to isomorphism and reordering of summands, as a finite direct sum of interval modules:

$$\mathbb{V} \;\cong\; \bigoplus_{i=1}^{m} \mathbb{F}_2[b_i, d_i).$$

The multiset $\{(b_i, d_i)\}_{i=1}^m$ is called the *barcode* of $\mathbb{V}$.

Proof sketch. Pick critical values $-\infty = t_0 < t_1 < \cdots < t_N < t_{N+1} = \infty$ such that $\mathbb{V}$ is constant on each open interval (t_i, t_{i+1}). Set $W_i := V_{t_i^+}$ (the right-limit at t_i, well-defined by finite type) and $\phi_i := v_{t_i^+, t_{i+1}^+} : W_i \to W_{i+1}$. We get a finite zigzag (in fact straight) sequence of $\mathbb{F}_2$-vector spaces

$$W_0 \xrightarrow{\phi_0} W_1 \xrightarrow{\phi_1} \dots \xrightarrow{\phi_{N-1}} W_N \xrightarrow{\phi_N} W_{N+1},$$

with $W_0 = W_{N+1} = 0$. This is exactly a finitely-generated $\mathbb{F}_2[t]$-module (where t acts via the shifts ϕ_i); but more elementarily, one classifies such persistence modules by induction on N using the following lemma, whose proof is direct.

Pivot lemma. Suppose $\dim W_i = w_i$. Choose bases compatibly so that each ϕ_i is a matrix in reduced column-echelon form. Each basis vector of W_i either lies in the kernel of ϕ_i (giving a death at t_{i+1}, paired with its birth), or maps to a basis vector of W_{i+1} (extending the corresponding bar). New basis vectors of W_{i+1} not in the image of ϕ_i correspond to births at t_{i+1}. The intervals so produced give the claimed decomposition; uniqueness follows because $\dim V_t$ and the rank of $v_{s,t}$ for $s < t$ are isomorphism invariants and together they determine the multiset of intervals via inclusion–exclusion (Theorem 5.6 below). □

Theorem 5.5 (Pointwise finite-dimensional case; Crawley-Boevey [CB15])**.** For any pfd persistence module over a totally ordered set, there is a unique direct-sum decomposition into interval modules (now possibly with $-\infty$ or ∞ endpoints and with open/closed endpoint subtleties).

We need only the finite-type case for finite filtrations of images, so Theorem 5.4 is sufficient. Theorem 5.5 is cited for completeness.

Theorem 5.6 (Rank function determines the barcode)**.** Two pfd persistence modules of finite type are isomorphic iff they have the same *rank function* $r(s,t) := \dim \operatorname{im} v_{s,t}$ for all $s \le t$.

Proof. For an interval module $\mathbb{F}_2[b,d)$, the rank function is $r(s,t) = \mathbf{1}_{s,t \in [b,d)}$. Ranks are additive over direct sums, so the rank function of $\bigoplus_i \mathbb{F}_2[b_i, d_i)$ is the multiset-cardinality function

$$r(s,t) \;=\; \#\{i : b_i \le s \le t < d_i\}.$$

By Möbius inversion on the half-plane $\{(s,t) : s \le t\}$ (or directly by inclusion–exclusion), r determines the multiset of pairs (b_i, d_i). □

5.2 Persistence diagram and barcode

Definition 5.7 (Persistence diagram)**.** The *persistence diagram* of a pfd persistence module $\mathbb{V}$ with finite-type barcode $\{(b_i, d_i)\}$ is the multiset

$$\mathrm{Dgm}(\mathbb{V}) \;:=\; \big\{(b_i, d_i)\big\}_i \;\cup\; \Delta_\infty,$$

where Δ_∞ denotes the diagonal $\{(x,x) : x \in \mathbb{R}\}$ taken with infinite multiplicity (technical convenience for matching in the bottleneck distance).

Remark 5.8 (Birth–death geometric meaning)**.** A point $(b,d) \in \mathrm{Dgm}(H_k(X_\bullet))$ encodes a k-dimensional topological feature that is born at value b (the first time a k-cycle representing it exists) and dies at value d (the first time that cycle becomes a boundary). The *lifetime* $d - b$ is the *persistence*; long-lived features sit far from the diagonal, short-lived ones near it.

What the studio renders. The persistence-diagram panel plots

- the points of $\mathrm{Dgm}(H_0(X_\bullet))$ in **cyan**;
- the points of $\mathrm{Dgm}(H_1(X_\bullet))$ in **magenta**;
- a glow halo around each point of radius $\propto \sqrt{d-b}$.

Counting points (b, d) with $b \le t < d$ yields $\beta_k(X_t)$ exactly. This count feeds the live indicator at the top of the diagram panel and the Betti curve panel.

6 Computation: union–find with the elder rule

The studio computes $\mathrm{Dgm}(H_0(X_\bullet))$ in near-linear time using union–find.

Definition 6.1 (Union–find structure)**.** A *disjoint-set* (or union–find) data structure on a finite set S maintains a partition of S. It supports

- MakeSet(x): create the singleton $\{x\}$;
- Find(x): return a canonical representative of the class containing x;
- Union(x, y): merge the classes of x and y.

With union-by-rank and path compression, any sequence of m operations on n elements runs in amortised $O(m\,\alpha(n))$ time, where α is the inverse Ackermann function [Tar75].

Algorithm 1 Sublevel H_0 persistence via union–find with the elder rule.

Sort pixels $p_1, \dots, p_N$ so that $f(p_1) < f(p_2) < \cdots < f(p_N)$ (genericity by tie-break).
Initialise an empty union–find structure U.
Initialise an empty multiset Dgm of pairs.
for $k = 1, \dots, N$ **do**
    $v \leftarrow f(p_k)$; activate p_k; MakeSet(p_k), record birth-site $b(p_k) \leftarrow p_k$.
    for each 4-neighbour q of p_k that is already active **do**
        $r_p \leftarrow$ Find(p_k); $r_q \leftarrow$ Find(q).
        if $r_p \neq r_q$ **then**
            Let $r^\star$ be the class with the *smaller* birth value (the *elder*);
              let $r^\dagger$ be the other class (the *younger*).
            Append $(f(b(r^\dagger)), v)$ to Dgm. ▷ younger dies at the merge value
            Union$(r^\star, r^\dagger)$, retaining $b(r^\star)$ as the new birth-site.
        end if
    end for
end for
Append $(f(b(r_\infty)), +\infty)$ to Dgm for the one essential class.
return Dgm

Theorem 6.2 (Elder-rule correctness)**.** For a sublevel-set filtration $\{X_t\}$ of the cubical pixel grid with generic f, Algorithm 1 outputs exactly the persistence diagram $\mathrm{Dgm}(H_0(X_\bullet))$, in $O(N\,\alpha(N))$ time after the initial sort.

Proof. By Lemma 2.3 the 1-skeleton of the cubical sublevel set X_t is the pixel graph $G_t = (\Omega_{\le t}, E_t)$, where E_t is the set of 4-adjacencies between pixels of $\Omega_{\le t}$. The 0-th homology of

a cell complex depends only on its 1-skeleton (attaching 2-cells adds no vertices and does not change which vertices are joined by paths), so $H_0(X_t) \cong H_0(G_t)$ for all t, naturally in t, and the persistence modules are isomorphic.

It therefore suffices to verify that Algorithm 1 computes $\mathrm{Dgm}(H_0(G_\bullet))$. We induct on the number of pixels processed. Let $G^{(k)}$ denote the graph after step k of the loop.

Base. $G^{(0)} = \emptyset$, $\beta_0(G^{(0)}) = 0$, $\mathrm{Dgm} = \emptyset$. Trivially correct.

Inductive step. Assume the algorithm correctly maintains Dgm at the start of iteration k such that the multiset of *closed* pairs in Dgm plus the currently active classes (one open pair per class) accounts for exactly $\beta_0(G^{(k-1)})$ live features at the present filtration value $f(p_{k-1})^+$; that is, one "open" pair per current component, born at the minimum value in that component.

At iteration k we activate p_k at value $v = f(p_k)$, increasing $\beta_0(G^{(k-1)})$ by 1 (a new isolated vertex) before any merges. The new class has birth v. We now process the active neighbours of p_k in any order:

- If p_k has no active neighbour, p_k remains an isolated component and $\beta_0(G^{(k)}) = \beta_0(G^{(k-1)}) + 1$. The new pair $(v, ?)$ is correctly opened.
- If p_k merges $r \geq 1$ previously distinct components, $\beta_0(G^{(k)}) = \beta_0(G^{(k-1)}) - r + 1$: we lose r components (the r old ones plus the freshly created $\{p_k\}$, total $r+1$) and gain one (their union), net change $-r$.

After the loop body for iteration k, the algorithm must close exactly r pairs at value v. The *elder rule* dictates that the r pairs closed are the r pairs with the largest birth values; the unique surviving open pair belongs to the component whose birth value is the smallest of the $r+1$ involved. This is exactly what lines 9–12 implement: each iteration of the inner loop merges *one* pair of classes, closing the younger and keeping the elder; after r such merges (one per neighbouring class), all $r+1$ involved classes are merged into one with the elder birth.

To see that the elder rule is the correct rule rather than an ad-hoc choice: by Theorem 5.6, the persistence diagram of $H_0(G_\bullet)$ is determined by the rank function $r(s,t) = \dim \mathrm{im}\big(H_0(G_s) \to H_0(G_t)\big)$. A class born at time b in G_s contributes to the rank at (s,t) iff it has a representative still alive at time t, i.e. it has not been absorbed by an *older* class by time t. Inducting on the number of merges, the unique consistent assignment of births/deaths is the one that always kills the younger class.

For the complexity: each pixel triggers $O(1)$ union–find operations, totalling $O(N)$ operations on $O(N)$ elements; amortised time is $O(N\alpha(N))$.

For the one remaining open pair at the end: G is connected (eventually) and $\beta_0(X) = 1$, so a single class survives. Line 15 outputs the pair $(f(b(r_\infty)), +\infty)$. (The studio replaces $+\infty$ by $\max f$ for display, which doesn't change the multiset structurally.) □

7 Alexander duality and H_1 for images

Theorem 7.1 (Alexander duality, [Hat02] §3.3)**.** Let A be a compact, locally contractible, nonempty subspace of S^n. For all $0 \leq k \leq n-1$, there is an isomorphism

$$\widetilde{H}_k(S^n \setminus A; \mathbb{F}_2) \;\cong\; \widetilde{H}^{n-k-1}(A; \mathbb{F}_2),$$

where $\widetilde{H}$ denotes reduced (co)homology. Over a field, $\widetilde{H}^{n-k-1}(A;\mathbb{F}_2) \cong \widetilde{H}_{n-k-1}(A;\mathbb{F}_2)$ by the universal coefficient theorem.

Proof idea, see [Hat02, Thm. 3.44]. Combine Lefschetz duality (for the compact manifold-with-boundary $S^n \setminus \mathrm{int}(N)$, where N is a regular neighbourhood of A) with excision and the fact that A deformation-retracts inside N. Each step is standard Čech-cohomological / cellular machinery. □

We need the case $n = 2,\ k = 0$. Compactify the plane to $S^2 = \mathbb{R}^2 \cup \{\infty\}$; for any finite cubical complex $A = X_t \subset \mathbb{R}^2$, the complement $S^2 \setminus A$ is open and locally contractible. Over $\mathbb{F}_2$,

$$\widetilde{H}_0(S^2 \setminus X_t; \mathbb{F}_2) \;\cong\; \widetilde{H}^1(X_t; \mathbb{F}_2) \;\cong\; \widetilde{H}_1(X_t; \mathbb{F}_2).$$

Proposition 7.2 (H_1 via bounded complement components)**.** For a cubical sublevel set $X_t \subset R \subset \mathbb{R}^2$,

$$\beta_1(X_t) \;=\; \#\big\{\text{bounded connected components of } \mathbb{R}^2 \setminus X_t\big\}.$$

Proof. $\widetilde{H}_0(S^2 \setminus X_t) = H_0(S^2 \setminus X_t) - 1$ (reduced homology subtracts one for the augmentation). Connected components of $S^2 \setminus X_t$ are connected components of $\mathbb{R}^2 \setminus X_t$ except that all the *unbounded* components of $\mathbb{R}^2 \setminus X_t$ collapse to a single component in S^2 via the point at infinity. For a bounded X_t inside the working rectangle, $\mathbb{R}^2 \setminus X_t$ has exactly one unbounded component (the "outside"), plus some bounded components (the holes). Hence

$$\#\text{components}(S^2 \setminus X_t) = 1 + \#\text{bounded components}(\mathbb{R}^2 \setminus X_t),$$

and $\widetilde{H}_0(S^2 \setminus X_t) = \#$bounded components. By Theorem 7.1 this equals $\widetilde{H}_1(X_t)$, and for the present spaces $\widetilde{H}_1 = H_1$ since they are connected enough that the reduction shift is trivial in degree 1. □

Proposition 7.3 (H_1 via superlevel pixel components)**.** Let $\widetilde{G}_t$ be the graph whose vertices are the superlevel pixels $\Omega_{>t}$ together with one extra vertex ∞, with an edge between two superlevel pixels iff they are 8-adjacent (king-move neighbours), and an edge between ∞ and every superlevel pixel on the border of Ω. Then the components of $\mathbb{R}^2 \setminus X_t$ correspond bijectively to the components of $\widetilde{G}_t$, the unbounded one to the component of ∞, so

$$\beta_1(X_t) \;=\; \#\big\{\text{components of } \widetilde{G}_t\big\} - 1 \;=\; \#\big\{\text{8-components of } \Omega_{>t} \text{ without border pixels}\big\}.$$

Proof. Let Q be a closed unit square of the grid with corners $a, b, c, d \in \Omega$. By Lemma 2.3, $Q \cap X_t$ is all of Q if the four corners are sublevel; otherwise it is the set of sublevel corners together with the sides of Q whose two endpoints are sublevel. Running through the cases (one, two adjacent, two diagonal or three sublevel corners), $Q \setminus X_t$ is then nonempty and path-connected, and it contains every superlevel corner of Q. Two superlevel pixels lie in a common unit square iff they are 8-adjacent, so 8-adjacent superlevel pixels lie in the same component of $R \setminus X_t$. Conversely, every point of $R \setminus X_t$ lies in some $Q \setminus X_t$ and is joined there to a superlevel corner, and a path in $R \setminus X_t$ can only pass from one unit square to the next through a point of a shared side that is not in X_t, that is, a side with a superlevel endpoint, which is a corner of both squares. Hence the components of $R \setminus X_t$ are the 8-components of $\Omega_{>t}$.

The region $\mathbb{R}^2 \setminus R$ is connected and unbounded. It meets the closure of a component of $R \setminus X_t$ exactly when that component reaches a point of ∂R outside X_t; such a point lies on a border side with a superlevel endpoint, which is a border pixel. Gluing $\mathbb{R}^2 \setminus R$ to the components that contain a border pixel is what the vertex ∞ does in $\widetilde{G}_t$. The bounded components of $\mathbb{R}^2 \setminus X_t$ are the remaining ones, and Prop. 7.2 gives the formula. □

The H_1 sweep. Prop. 7.3 reduces H_1 to H_0 of the graphs $\widetilde{G}_t$, which grow as t *decreases*. Persephone therefore runs Algorithm 1 a second time, with three changes:

1. pixels are processed in *descending* order of f (the reverse of the tie-broken ascending order), and each new pixel is joined to its already active 8-*neighbours*;
2. the vertex ∞ is active from the start and is joined to every border pixel when that pixel enters;

3. the elder of two classes is the one whose birth pixel was processed *first* in the sweep, with ∞ older than every pixel. In the descending sweep this is the class with the *larger* birth value; keeping the class with the smaller value, as in the ascending sweep, would kill the wrong one.

With these changes the proof of Theorem 6.2, applied to $-f$ on the graphs $\widetilde{G}_t$, shows that the sweep outputs the H_0 diagram of the superlevel filtration. The class of ∞ never dies and is discarded; it is the unbounded complement component, not a hole. Every other class is born at a local maximum b and dies at the value $d < b$ at which it merges with an older class or reaches the border. It is a bounded component of $\mathbb{R}^2 \setminus X_t$ exactly for $t \in [d, b)$, so it is reported as the H_1 pair (d, b) of the sublevel filtration. Alexander duality is natural with respect to inclusions, so the maps $H_1(X_s) \to H_1(X_t)$ are dual to the maps on reduced H_0 of the complements; the ranks of the two persistence modules therefore agree and the diagrams coincide, not only the Betti numbers. Both sweeps run in $O(N\,\alpha(N))$ after the sort.

Using 4-adjacency in this sweep as well would be incorrect, and not only on rare inputs: two superlevel pixels that touch diagonally would count as separate holes although the complement connects them through the corner (a diagonal ring of sublevel pixels, or a 2×2 checkerboard, would report holes that do not exist), and every peak that reaches the border would count as a hole without the vertex ∞. The repository's test suite compares both diagrams with a brute-force computation of the components and holes of every sublevel set, on hand-made diagonal cases and on random images.

8 Persistence landscapes

Bubenik [Bub15] introduced a 1-parameter family of real-valued functions encoding the persistence diagram, designed for downstream statistical use.

Definition 8.1 (Tent function)**.** For a pair $(b, d) \in \mathbb{R}^2$ with $b < d$, the *tent function* is

$$\Lambda_{(b,d)}(t) \;:=\; \max\big\{\,0,\;\min(t-b,\;d-t)\,\big\}, \qquad t \in \mathbb{R}.$$

Lemma 8.2 (Tent properties)**.** $\Lambda_{(b,d)}$ is supported on $[b, d]$, 1-Lipschitz, piecewise linear, and reaches its maximum value $(d-b)/2$ at $t = (b+d)/2$.

Proof. $\Lambda_{(b,d)}(t) = 0$ outside $[b, d]$ because for $t < b$, $t - b < 0$, and for $t > d$, $d - t < 0$, so the inner min is negative and the outer max yields 0. Inside $[b, d]$, $\Lambda_{(b,d)}(t) = \min(t-b, d-t)$. Each of $t - b$ and $d - t$ has slope ± 1, so their min is 1-Lipschitz. Outside $[b, d]$ the function is constantly 0, also 1-Lipschitz. The maximum is attained where $t-b = d-t$, i.e. at $t = (b+d)/2$ with value $(d-b)/2$. □

Definition 8.3 (Persistence landscape)**.** Let $\mathrm{Dgm} = \{(b_i, d_i)\}_{i \in I}$ be a persistence diagram (with possibly infinite multiplicity diagonal, but only finitely many off-diagonal pairs). Its *k-th landscape function* is

$$\lambda_k(t) \;:=\; \mathrm{kmax}_{i \in I}\, \Lambda_{(b_i,d_i)}(t),$$

the k-th largest value among the tent heights at t, with the convention $\lambda_k(t) = 0$ if fewer than k pairs have $\Lambda_{(b_i,d_i)}(t) > 0$.

Lemma 8.4 (k-max is 1-Lipschitz)**.** Let $g_1, \dots, g_n : \mathbb{R} \to \mathbb{R}$ be 1-Lipschitz functions and let $g_{(k)}(t)$ denote the k-th largest value of $g_1(t), \dots, g_n(t)$. Then $g_{(k)}$ is 1-Lipschitz.

Proof. Fix $s, t \in \mathbb{R}$ and let $A_s \subseteq \{1, \dots, n\}$ be the indices of the k largest $g_i(s)$. Then $g_{(k)}(s) = \min_{i \in A_s} g_i(s)$. By 1-Lipschitz of each g_i, $g_i(t) \geq g_i(s) - |s - t|$ for all i. Among the k indices in A_s, the smallest value at t is at most $g_{(k)}(t)$ (taking the k-th largest of all n values), and at least $g_{(k)}(s) - |s - t|$. Hence $g_{(k)}(t) \geq g_{(k)}(s) - |s - t|$. The symmetric argument gives $g_{(k)}(s) \geq g_{(k)}(t) - |s - t|$, so $|g_{(k)}(s) - g_{(k)}(t)| \leq |s - t|$. □

Proposition 8.5 (Landscape properties)**.** For every persistence diagram Dgm (finitely many off-diagonal pairs) and every $k \geq 1$:

(i) λ_k is 1-Lipschitz, hence continuous.

(ii) $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ pointwise.

(iii) λ_k has support contained in $\bigcup_i [b_i, d_i]$.

(iv) The map $\mathrm{Dgm} \mapsto (\lambda_k)_{k \geq 1}$ is injective: the landscape uniquely determines the diagram.

Proof. **(i)** Each tent $\Lambda_{(b_i, d_i)}$ is 1-Lipschitz (Lemma 8.2); the k-th largest of finitely many 1-Lipschitz functions is 1-Lipschitz (Lemma 8.4). Constant zero diagonal tents contribute nothing.

(ii) By definition $\lambda_k(t)$ is the k-th largest among $\{\Lambda_{(b_i, d_i)}(t)\}_i$, monotone decreasing in k. All values are ≥ 0 since each tent is ≥ 0.

(iii) If $t \notin \bigcup_i [b_i, d_i]$ then $\Lambda_{(b_i, d_i)}(t) = 0$ for all i, so $\lambda_k(t) = 0$.

(iv) Sum and difference of landscapes: $\sum_{k \geq 1} \lambda_k(t) = \sum_i \Lambda_{(b_i, d_i)}(t)$, since the sum of all tent values at t equals the sum of all k-th largest values (just a reordering). The latter is a sum of finitely many tents, hence determines the diagram up to multi-set (by analyzing the slope discontinuities; each tent has unique apex $(b + d)/2$ and slope-change points at b and d, so the multi-set of break points and apexes can be read off). A standard argument (cf. [Bub15, Thm. 12]) shows that the sequence $(\lambda_k)_k$ further refines this enough to be injective on multisets.

(For completeness: the sequence (λ_k) determines $(\lambda_1 + \lambda_2 + \dots)$ and from this sum the multi-set of pairs is recovered as in the previous paragraph.) □

The studio computes $\lambda_1, \dots, \lambda_{14}$ on a uniform sample of 200 values of t and lifts them into $\mathbb{R}^3$: the x-axis is t, the y-axis is $\lambda_k(t)$, the z-axis is k. Adjacent levels are stitched with translucent additive triangles, producing the glowing wireframe terrain.

9 Stability

A core theorem of TDA, and the reason persistence is a useful descriptor under noise, states that persistence diagrams are Lipschitz-stable under perturbation of the input function.

Definition 9.1 (Bottleneck distance)**.** For two persistence diagrams $\mathrm{Dgm}_1, \mathrm{Dgm}_2$ (each augmented by the diagonal with infinite multiplicity), the *bottleneck distance* is

$$d_B(\mathrm{Dgm}_1, \mathrm{Dgm}_2) := \inf_{\eta} \sup_{x \in \mathrm{Dgm}_1} \|x - \eta(x)\|_\infty,$$

where the infimum is over bijections $\eta : \mathrm{Dgm}_1 \to \mathrm{Dgm}_2$.

Definition 9.2 (ε-interleaving)**.** Two persistence modules $\mathbb{U} = (U_t, u_{s,t})$ and $\mathbb{V} = (V_t, v_{s,t})$ are *ε-interleaved* ($\varepsilon \geq 0$) if there exist families of linear maps

$$\phi_t : U_t \to V_{t+\varepsilon}, \qquad \psi_t : V_t \to U_{t+\varepsilon}, \qquad t \in \mathbb{R},$$

that are compatible with the module structure and satisfy

$$\psi_{t+\varepsilon} \circ \phi_t = u_{t, t+2\varepsilon}, \qquad \phi_{t+\varepsilon} \circ \psi_t = v_{t, t+2\varepsilon}.$$

The *interleaving distance* is

$$d_I(\mathbb{U}, \mathbb{V}) \;:=\; \inf\{\varepsilon \geq 0 : \mathbb{U} \text{ and } \mathbb{V} \text{ are } \varepsilon\text{-interleaved}\}.$$

Lemma 9.3 (Function bound implies interleaving)**.** Let $f, g : K \to \mathbb{R}$ be functions on a finite CW complex giving sublevel filtrations $X_t^f = f^{-1}(-\infty, t]$ and $X_t^g = g^{-1}(-\infty, t]$ (made into CW subcomplexes by the V-construction). Set $\varepsilon = \|f - g\|_\infty := \sup_{x \in K} |f(x) - g(x)|$. Then $X_t^f \subseteq X_{t+\varepsilon}^g$ and $X_t^g \subseteq X_{t+\varepsilon}^f$ for all t, hence $H_k(X_\bullet^f)$ and $H_k(X_\bullet^g)$ are ε-interleaved.

Proof. If $f(x) \leq t$, then $g(x) \leq f(x) + \varepsilon \leq t + \varepsilon$. Hence $X_t^f \subseteq X_{t+\varepsilon}^g$, and the inclusion induces $\phi_t : H_k(X_t^f) \to H_k(X_{t+\varepsilon}^g)$. The symmetric argument gives ψ_t. The interleaving identities $\psi_{t+\varepsilon} \circ \phi_t =$ inclusion-induced map $H_k(X_t^f) \to H_k(X_{t+2\varepsilon}^f)$ follow from the commutativity of inclusion squares. □

Theorem 9.4 (Algebraic stability; Chazal–Cohen-Steiner–Glisse–Guibas–Oudot [CCSGGO09])**.** For any two pfd persistence modules $\mathbb{U}, \mathbb{V}$ of finite type,

$$d_B(\mathrm{Dgm}(\mathbb{U}), \mathrm{Dgm}(\mathbb{V})) \;\leq\; d_I(\mathbb{U}, \mathbb{V}).$$

The reverse inequality also holds (isometry theorem; [Les15]), but the bound above is what we need.

Proof outline. The algebraic stability theorem is proved by exhibiting, for any ε-interleaving, an explicit matching between the bars of $\mathbb{U}$ and $\mathbb{V}$ with ℓ^∞ displacement at most ε. The cleanest argument uses the *induced matching* of Bauer–Lesnick [BL15]: given an interleaving, one constructs canonical maps $\phi_* : \mathrm{Dgm}(\mathbb{U}) \to \mathrm{Dgm}(\mathbb{V})$ (and vice versa) that move every off-diagonal point by at most ε in ℓ^∞. The induced matching is well-defined because pfd persistence modules over a totally ordered set decompose canonically into intervals (Theorem 5.5), and the interleaving maps respect this decomposition up to diagonal noise. The full proof is technical; see [BL15, CCSGGO09] for details. □

Theorem 9.5 (Persistence stability; Cohen-Steiner–Edelsbrunner–Harer [CSEH07])**.** Let $f, g : K \to \mathbb{R}$ be functions on a finite CW complex. Then for every degree k,

$$d_B\big(\mathrm{Dgm}(H_k(f)), \mathrm{Dgm}(H_k(g))\big) \;\leq\; \|f - g\|_\infty.$$

Proof. By Lemma 9.3, the persistence modules $H_k(X_\bullet^f)$ and $H_k(X_\bullet^g)$ are $\|f - g\|_\infty$-interleaved. By Theorem 9.4, their persistence diagrams have bottleneck distance at most $\|f - g\|_\infty$. □

Remark 9.6. For the image setting: a perturbation of pixel intensities by at most ε moves every point of the persistence diagram by at most ε in the ℓ^∞ sense. In particular, low-amplitude photographic noise produces only near-diagonal noise in the diagram, which the studio fades visually (small lifetime = small halo).

Corollary 9.7 (Landscape stability)**.** For any two pfd persistence diagrams arising from functions $f, g : K \to \mathbb{R}$, the persistence landscapes satisfy

$$\|\lambda_k^f - \lambda_k^g\|_\infty \;\leq\; \|f - g\|_\infty \qquad \text{for every } k \geq 1.$$

Proof. By Bubenik [Bub15, Thm. 16], the landscape distance is bounded by the bottleneck distance: $\|\lambda_k^f - \lambda_k^g\|_\infty \leq d_B(\mathrm{Dgm}(f), \mathrm{Dgm}(g))$. Combined with Theorem 9.5, this gives the claim. □

10 What every panel shows

This section maps each visible panel of the studio to the mathematical object it renders.

Sublevel-set canvas (centre)

A V-construction visualisation of the cubical filtration (Lemma 2.3). For the current t the shader draws:

- the sublevel set $\Omega_{\le t}$ in a *warm* (magma) palette;
- the superlevel set $\Omega_{>t}$ in a *cool* (viridis) palette, dimmed;
- a glowing tide front along the level set $\{p : f(p) = t\}$ whose band width is inversely proportional to $\|\nabla f\|$, so flat regions have a wide soft glow and steep edges a tight sharp glow.

Persistence diagram (top right)

The multiset $\mathrm{Dgm}(H_0) \sqcup \mathrm{Dgm}(H_1)$ (Definition 5.7) plotted in the (birth, death) plane, above the diagonal. Active features (birth $\le t <$ death) are drawn bright; expired features are dimmed. Counting points with $b \le t < d$ yields $\beta_k(X_t)$ by Theorem 5.6.

Persistence barcode (bottom centre)

Each pair $(b, d) \in \mathrm{Dgm}$ rendered as a horizontal bar $[b, d]$, sorted by birth. The vertical guideline marks the current t; bars intersecting it are alive at t.

Betti curves (centre, above the barcode)

The functions

$$\beta_0(t) \;=\; \#\{(b,d) \in \mathrm{Dgm}_0 : b \le t < d\}, \qquad \beta_1(t) \;=\; \#\{(b,d) \in \mathrm{Dgm}_1 : b \le t < d\},$$

sampled on a uniform grid in t. These are step functions that the renderer interpolates linearly. Their alternating sum is the live Euler characteristic from Theorem 4.7.

Persistence landscape (bottom right, 3D)

The stack $\{(\,t,\, \lambda_k(t),\, k\,)\}$ from Definition 8.3, $k = 1, \dots, 14$, sampled on a uniform 200-point grid. Layers are stitched into translucent triangle strips and rendered with additive bloom; an autorotating orbit camera reveals the topography. Proposition 8.5 guarantees continuity (smoothness of the rendered curves) and the monotone-decreasing ordering of layers.

Invariants strip (top right)

The live integers

$$\beta_0(t), \quad \beta_1(t), \quad \chi(t) = \beta_0(t) - \beta_1(t),$$

the total cardinality $|\mathrm{Dgm}|$, and the working raster size. The identity $\chi = \beta_0 - \beta_1$ is Theorem 4.7.

References

[BL15] U. Bauer, M. Lesnick. *Induced matchings and the algebraic stability of persistence barcodes.* J. Comput. Geom. 6 (2015), 162–191.

[Bub15] P. Bubenik. *Statistical topological data analysis using persistence landscapes.* J. Mach. Learn. Res. 16 (2015), 77–102.

[Car09] G. Carlsson. *Topology and data.* Bull. Amer. Math. Soc. 46 (2009), no. 2, 255–308.

[CCSGGO09] F. Chazal, D. Cohen-Steiner, M. Glisse, L. J. Guibas, S. Y. Oudot. *Proximity of persistence modules and their diagrams.* Proc. SoCG, 2009.

[CSEH07] D. Cohen-Steiner, H. Edelsbrunner, J. Harer. *Stability of persistence diagrams.* Discrete Comput. Geom. 37 (2007), 103–120.

[CB15] W. Crawley-Boevey. *Decomposition of pointwise finite-dimensional persistence modules.* J. Algebra Appl. 14 (2015), no. 5.

[EH10] H. Edelsbrunner, J. Harer. *Computational Topology: An Introduction.* AMS, 2010.

[Gud] The GUDHI Project. *Cubical complexes and persistent homology.* https://gudhi.inria.fr.

[Hat02] A. Hatcher. *Algebraic Topology.* Cambridge University Press, 2002.

[Les15] M. Lesnick. *The theory of the interleaving distance on multidimensional persistence modules.* Found. Comput. Math. 15 (2015), 613–650.

[Tar75] R. E. Tarjan. *Efficiency of a good but not linear set union algorithm.* J. ACM 22 (1975), 215–225.

[WCV12] H. Wagner, C. Chen, E. Vuçini. *Efficient computation of persistent homology for cubical data.* In *Topological Methods in Data Analysis and Visualization II*, Springer, 2012, 91–106.

[ZC05] A. Zomorodian, G. Carlsson. *Computing persistent homology.* Discrete Comput. Geom. 33 (2005), 249–274.
